MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.1: General functions and domains

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Function types and component functions

Unit 1 and Unit 2 studied maps of the form $\mathbf{x} \mapsto A\mathbf{x}$. We now allow more general rules. The output may be a scalar or a vector.
In Unit 2, a data matrix turned model coefficients into predictions. How well do those predictions match observed values, and how should we change the coefficients to improve the fit? A single number measuring the prediction error will give us a scalar-valued function to study throughout this unit.

Definition

A vector-valued map is usually written $F : \mathbb{R}^n \to \mathbb{R}^m$. A scalar-valued function is usually written $f : \mathbb{R}^n \to \mathbb{R}$. Any vector-valued map $F : \mathbb{R}^n \to \mathbb{R}^m$ has m component functions, or coordinate functions, the scalar-valued functions $F_1, \dots, F_m : \mathbb{R}^n \to \mathbb{R}$ such that

$$F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix}.$$

More generally, vector-valued and scalar-valued functions may only be defined on a subset of vectors in $\mathbb{R}^n$, rather than defined everywhere, so their domain may be a proper subset of $\mathbb{R}^n$.

In other words, a scalar-valued function takes in an input and outputs *scalars*, and a vector-valued function takes in an input and outputs *vectors* (recall Vector).

Activity: Vector length as a scalar-valued function (U3-LO1)

If $\mathbf{x}$ is a point in $\mathbb{R}^n$, recall the length of $\mathbf{x}$, denoted $\|\mathbf{x}\|$, from Length of a vector. The length of a vector defines a scalar-valued function from $\mathbb{R}^n$ to $\mathbb{R}$. Find two different vectors with the same length. What does this say about whether the output determines the input?

Activity: Component functions of a matrix map (U3-LO1)

Let A be an $m \times n$ matrix. The linear map (recall Linear map) $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\mathbf{x}) = T_A(\mathbf{x}) = A\mathbf{x}$ is a vector-valued map. Each component function is a row dot product.
If $\mathbf{a}_1, \ldots, \mathbf{a}_m$ are the rows of A, viewed as vectors in $\mathbb{R}^n$, and $T_1, \ldots, T_m$ are the component functions of T, then for each j, express the function T_j in terms of the row $\mathbf{a}_j$.

Activity: Component functions of a matrix map (U3-LO1)

Let A be an $m \times n$ matrix. The linear map (recall Linear map) $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\mathbf{x}) = T_A(\mathbf{x}) = A\mathbf{x}$ is a vector-valued map. Each component function is a row dot product.
Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{bmatrix},$$

which defines a linear map $T_A : \mathbb{R}^3 \to \mathbb{R}^2$. Compute its component functions for a general input $\mathbf{x}$, and its output for the specific input $\mathbf{x} = (3, -1, 2)$.

Activity: Corners are not enough (U3-LO1)

Unit 1 used the unit-square visualization for 2×2 matrices. For nonlinear maps, the corners of a square do not tell the whole story. Let $T, F : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

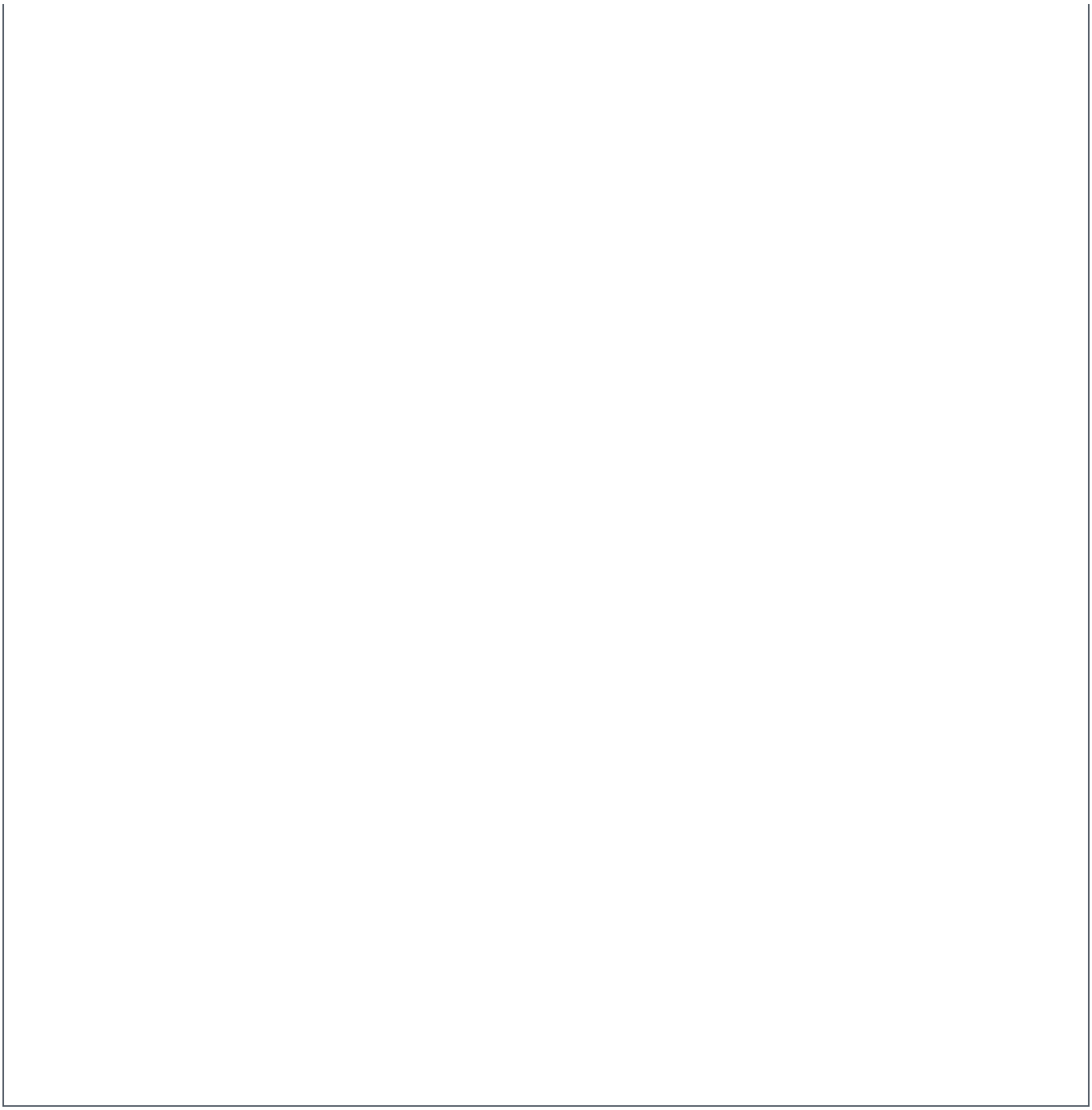
$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y \\ y \end{bmatrix}, \qquad F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

1. Compute the images of the four corners of the unit square under T.
2. Compute the images of the four corners of the unit square under F.

Activity: Corners are not enough (U3-LO1) (continued)

3. Compute the images of $\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ under both maps.

4. What does F do to a point $\begin{bmatrix} 0 \\ y \end{bmatrix}$ on the left edge of the unit square?



Note

A matrix sends grid lines to grid lines. A nonlinear map can bend grid lines. The square-grid visualization samples points and grid lines, not only the corners.

Activity: Reading a square-grid computation (U3-LO1)

The following code applies the nonlinear map

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}$$

to several input points stored as columns.
In NumPy, `**` denotes exponentiation; for example, `x**2` represents x^2.

```
import numpy as np

def F(P):
    x = P[0, :]
    y = P[1, :]
    return np.vstack([x + y**2, y])

P = np.array([
    [0.0, 1.0, 1.0, 0.0, 0.0],
    [0.0, 0.0, 1.0, 1.0, 0.5],
])

F(P)
```

Output:

```
array([[0.  , 1.  , 2.  , 1.  , 0.25],
       [0.  , 0.  , 1.  , 1.  , 0.5 ]])
```

1. What points are stored in the first four columns of P?
2. What point is stored in the last column of P?
3. What output corresponds to the last column?
4. Why do the first four columns not detect the difference between F and the shear $T(x, y) = (x + y, y)$?

Activity: Reading a square-grid computation (U3-LO1) (continued)

5. Why is a grid more informative than the four corners?

Note: Reading square-grid diagrams

A square-grid diagram keeps track of input points and output points at

the same time. The color identifies the original input point. The grid lines show how nearby input points move.
For a matrix map, grid lines stay straight and the whole grid is controlled by one matrix. For a smooth nonlinear map, grid lines may bend.
The purpose of the diagram is not to replace computation. It shows what the computation is doing.

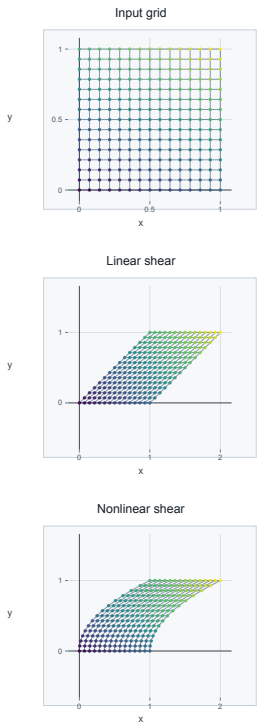


Figure. Square-grid visualization for $T(x, y) = (x + y, y)$ and $F(x, y) = (x + y^2, y)$. The color identifies the original input point. The two maps agree on the four corners, but the nonlinear map bends grid lines.

Activity: From predictions to a scalar error function (U3-LO1, U2-LO1)

As in Two coefficient choices, same predictions (U2-LO1, U2-LO6), the input to a prediction map is a vector of model coefficients. Consider three observations with two numerical features each:

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Activity: From predictions to a scalar error function (U3-LO1, U2-LO1) (continued)

Each row of X gives one observation's features, and the corresponding entry of $\mathbf{y}$ is its observed response. Both X and $\mathbf{y}$ stay fixed. The two adjustable weights are $\mathbf{w} = (w_1, w_2)$.
The prediction vector and the error vector, also called the residual, are

$$P(\mathbf{w}) = X\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_1 + w_2 \end{bmatrix}, \qquad R(\mathbf{w}) = X\mathbf{w} - \mathbf{y}.$$

Using the vector length from Length of a vector, we summarize these three errors by the squared-error loss

$$\mathscr{L}(\mathbf{w}) = \|X\mathbf{w} - \mathbf{y}\|^2 = (w_1 - 1)^2 + (w_2 - 2)^2 + (w_1 + w_2 - 2)^2.$$

A smaller loss means a better fit to these observations.

1. Give the domains and co-domains of P, R, and $\mathscr{L}$. Which are vector-valued, and which is scalar-valued? Which map is linear, and which is affine?
2. Compute the predictions, residuals, and losses at $\mathbf{w}_0 = (1, 1)$ and $\mathbf{b} = (0, 2)$. Does equal loss imply equal predictions?
3. Could any choice of weights give loss zero? Translate this question into a linear system and use Unit 2's row-reduction method.

The notation $F : \mathbb{R}^n \to \mathbb{R}^m$ includes several cases. If $m = 1$, the output is scalar-valued. If $n = 1$ and $m > 1$, the map traces a curve. If $F(\mathbf{x}) = A\mathbf{x}$, the map is linear. If $F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, the map is affine.

Domains, ranges, graphs, and level curves

We encountered the domain of a function in Linear and affine maps. We now give a more formal definition and introduce the range.

Definition: Domain and range

The domain of a function is the set of all inputs for that function, and the range is the set of all possible outputs. More formally, if a function f takes inputs from a set X and outputs elements in a set Y, then the domain of f is the set X, and the range is the set of all $y \in Y$ for which $y = f(x)$ for some $x \in X$, which we might write in set notation as $\{y \in Y : y = f(x) \text{ for some } x \in X\}$.

In calculus, we often define the domain of a function f *implicitly*, by defining f in terms of an expression, and then letting the domain of f be the set of all values which, when substituted into that expression, lead to a well-defined quantity. For instance, we might discuss 'the function' $f(x, y) = x/y$, by which we mean the function with domain $\{(x, y) \in \mathbb{R}^2 : y \neq 0\}$ and range $\mathbb{R}$.

The range contains only outputs actually attained by the function, while the co-domain, defined in Linear and affine maps, is the specified output set and may be larger than the range.

For scalar-valued functions of two variables, domains and level curves are subsets of the plane. These pictures help us see where a function is defined and how its outputs change.

Activity: Sketching domains (U3-LO1)

Sketch the domains of the following functions:
$f(x, y) = \ln(y - x^2) + \sqrt{x}$.

Activity: Sketching domains (U3-LO1)

Sketch the domains of the following functions:
$f(x, y) = e^x \sin y$

Activity: Sketching domains (U3-LO1)

Sketch the domains of the following functions:

$f(x, y) = \frac{\sqrt{x}}{\sqrt{y}}$

Definition: The Graph of a Function

The graph of a function f is the set of all pairs $(x, f(x))$, with x in the domain of f.

When f is a scalar-function of two variables, the graph of f is a set of triples $(x, y, f(x, y))$, and thus a subset of $\mathbb{R}^3$, and can therefore be visualized as a surface. See the figure for a picture of the graph of the function $f(x, y) = xe^{-x^2-y^2}$.

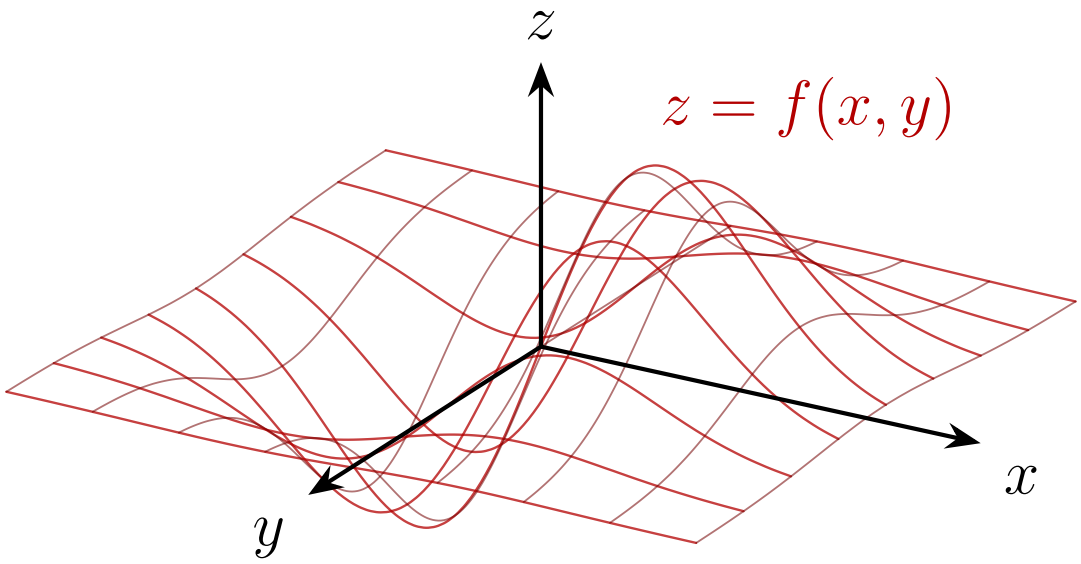


Figure. A graph of the function $f(x, y) = xe^{-x^2-y^2}$.

Definition: Level Curves and Contour Plots

A level curve of a scalar function f of two variables is the curve consisting of pairs (x, y) satisfying the equation $f(x, y) = c$, for some $c \in \mathbb{R}$. A contour plot is a sketch in $\mathbb{R}^2$ depicting level curves of a function f for many different values of c.

The different values of c which are chosen to be depicted in a contour plot are often (but not always) obtained by varying c by a common difference, obtaining 'consecutive' level sets.

Activity: Sketching level curves (U3-LO1)

Sketch a contour plot for the function

$$f(x, y) = e^{x^2+y^2}$$

by sketching the level curves $f(x, y) = c$ with $c = 1$, $c = e$, $c = e^4$, and $c = e^9$.

Activity: Equal-error curves in parameter space (U3-LO1)

Return to the loss from From predictions to a scalar error function (U3-LO1, U2-LO1):

$$\mathscr{L}(w_1, w_2) = (w_1 - 1)^2 + (w_2 - 2)^2 + (w_1 + w_2 - 2)^2.$$

1. Describe the coordinates of a point on the graph of $\mathscr{L}$. Which coordinates represent model weights, and which represents prediction error?
2. Write the equation of the level curve $\mathscr{L} = 1$. Locate $(1, 1)$ and $(0, 2)$ on this curve, and give the corresponding points on the graph. What would moving along this level curve preserve?
3. Can a level curve have a negative loss value? Is the level set $\mathscr{L} = 0$ nonempty for these data?

Balls, limits, and continuity

This short section supplies local language used in derivative and optimization statements: balls, limits, continuity, interior points, boundary points, closed sets, and bounded sets.
Recall the definition of a limit from single-variable calculus. For a function $f : \mathbb{R} \to \mathbb{R}$ and $a, b \in \mathbb{R}$, the equation

$$\lim_{x \to a} f(x) = b$$

holds if for each $\varepsilon > 0$, there is $\delta > 0$ so that if $0 < |x - a| < \delta$, then $|f(x) - b| < \varepsilon$. We can interpret this statement geometrically as follows: for any open interval J centered at b, there is an open interval I centered at a so that f maps points in I, excluding a itself, into J.
In $\mathbb{R}^2$, a ball is a disk.

Definition: Disks in $\mathbb{R}^2$

Consider a point $(a, b) \in \mathbb{R}^2$. The δ-disk centered at point (a, b) is the region of $\mathbb{R}^2$ bounded by the circle of radius δ centered at (a, b), i.e., the set

$$\left\{(x, y) \in \mathbb{R}^2 \, : \, \sqrt{(x-a)^2 + (y-b)^2} < \delta\right\}.$$

See the figure for a depiction of a δ disk.

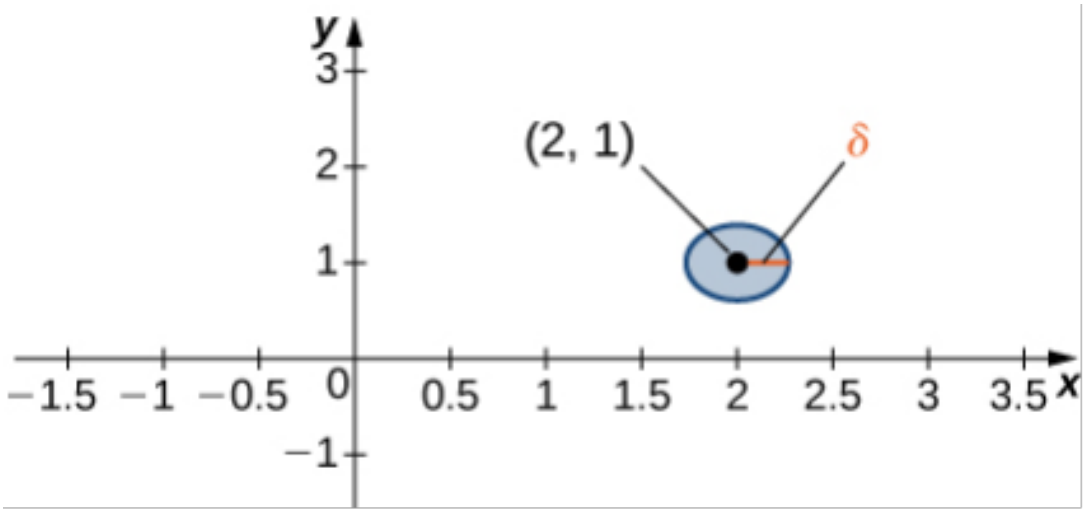


Figure. A δ-disk centered around the point $(2, 1)$. Figure 4.14 from Edwin "Jed" Herman and Gilbert Strang, *Calculus Volume 3*, OpenStax, © 2018 Rice University, licensed under CC BY-NC-SA 4.0; source: OpenStax Figure 4.14.

We now define the limit of a scalar-valued function f of two variables. For this definition, the equation

$$\lim_{(x,y)\to(a,b)} f(x, y) = L,$$

will then hold precisely when, for each open interval I centered at L, there is a δ disk centered at (a, b) so that all points in this disk which are also contained in the domain of f, excluding the point (a, b) itself, are mapped by f into I.

Definition: Limits of Scalar-Valued Functions of Two Variables

Consider a scalar-valued function $f(x, y)$ of two variables.

The limit of f as (x, y) approaches (a, b) is equal to L, written

$$\lim_{(x,y)\to(a,b)} f(x, y) = L,$$

if for each $\varepsilon > 0$, there is $\delta > 0$ so that for any points in the domain of f with

$$0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta,$$

the inequality $|f(x,y) - L| < \varepsilon$ holds.

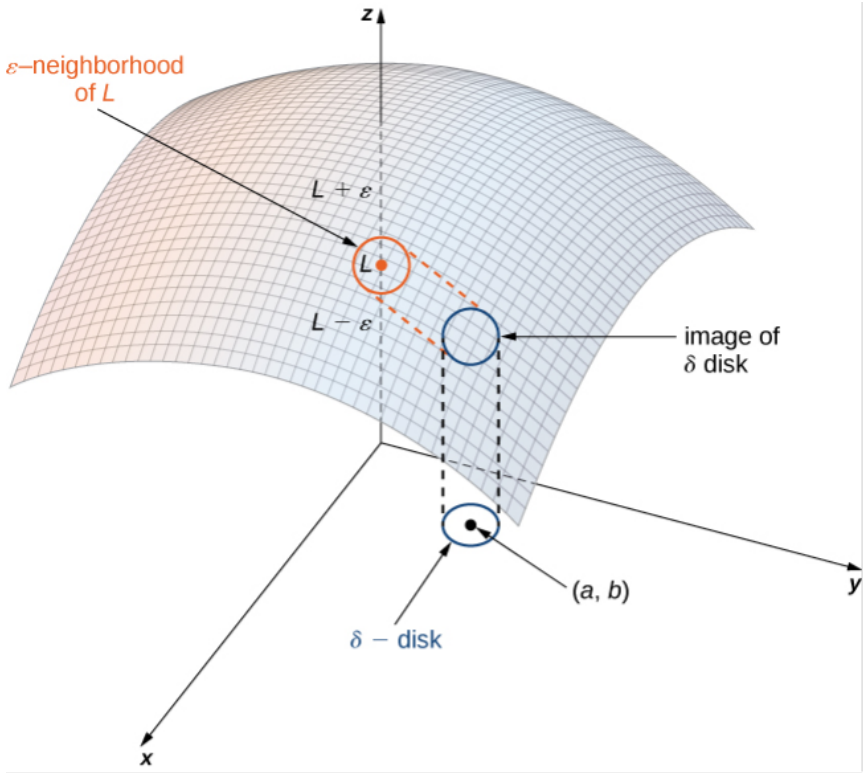


Figure. Epsilon-delta diagram for a two-variable limit on a surface.

Using vector-notation (recalling Length of a vector), we can write a δ-disk centered at a point $\mathbf{x}_0 \in \mathbb{R}^2$ as the set of points $\mathbf{x} \in \mathbb{R}^2$ so that

$$\{\mathbf{x} \in \mathbb{R}^2 \,:\, \|\mathbf{x} - \mathbf{x}_0\| < \delta\}.$$

This motivates the higher dimensional variant of a δ disk: a *δ-ball*.

Definition: Balls in $\mathbb{R}^n$

For $\mathbf{x}_0 \in \mathbb{R}^n$, the δ-ball centered at $\mathbf{x}_0$ is the set

$$\{\mathbf{x} \in \mathbb{R}^n \,:\, \|\mathbf{x} - \mathbf{x}_0\| < \delta\}.$$

In $\mathbb{R}^3$ a δ-ball centered at a point (a,b,c) is precisely the set of points satisfying the equation

$$(x-a)^2 + (y-b)^2 + (z-c)^2 < \delta^2.$$

It is the region in $\mathbb{R}^3$ bounded by a sphere of radius δ.
By replacing disks with balls, we obtain the definition of a limit of a general scalar-valued function.

Definition: The Limit of a Scalar-Valued Function

Consider a scalar-valued function f.

The limit of f as $\mathbf{x}$ approaches $\mathbf{x}_0$ is equal to L, written

$$\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = L,$$

if for each $\varepsilon > 0$, there is $\delta > 0$ so that for any points in the domain of f with $0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta$, the inequality $|f(\mathbf{x}) - L| < \varepsilon$ holds.

Remark

In order for the limit of f, when it exists, to be unique, the point $\mathbf{x}_0$ must have the property that the domain of f contains a point other than $\mathbf{x}_0$ in any δ-ball centered at $\mathbf{x}_0$. This is only a small issue, since every point in the domain of any of the functions that naturally occur in calculus have this property.

Recall from the calculus of a single variable that a function $f : \mathbb{R} \to \mathbb{R}$ is continuous at a point a when $\lim_{x\to a} f(x) = f(a)$, i.e., when the limit on the left hand side exists and is equal to the right hand side. The continuity of functions of several variables is defined in exactly the same way.

Definition

A scalar-valued function f is continuous at a point $\mathbf{x}_0$ in its domain if the limit of f as $\mathbf{x} \to \mathbf{x}_0$ exists, and

$$\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0).$$

Interior, boundary, and closed sets

This language lets us distinguish points inside a set from points on its boundary.

Definition: Interior, boundary, and closed sets

Let S be a subset of $\mathbb{R}^n$. Then a point $\mathbf{x}_0 \in S$ is an interior point if there is a δ-ball centered at $\mathbf{x}_0$ completely contained in S. A point $\mathbf{x}_0 \in \mathbb{R}^n$ is a boundary point if every ball centered at $\mathbf{x}_0$ contains at least one point of S and at least one point outside S. The set of all boundary points of a set S is called the boundary of the set S.

Consider the figure for an illustration of the example of an interior point and boundary point.

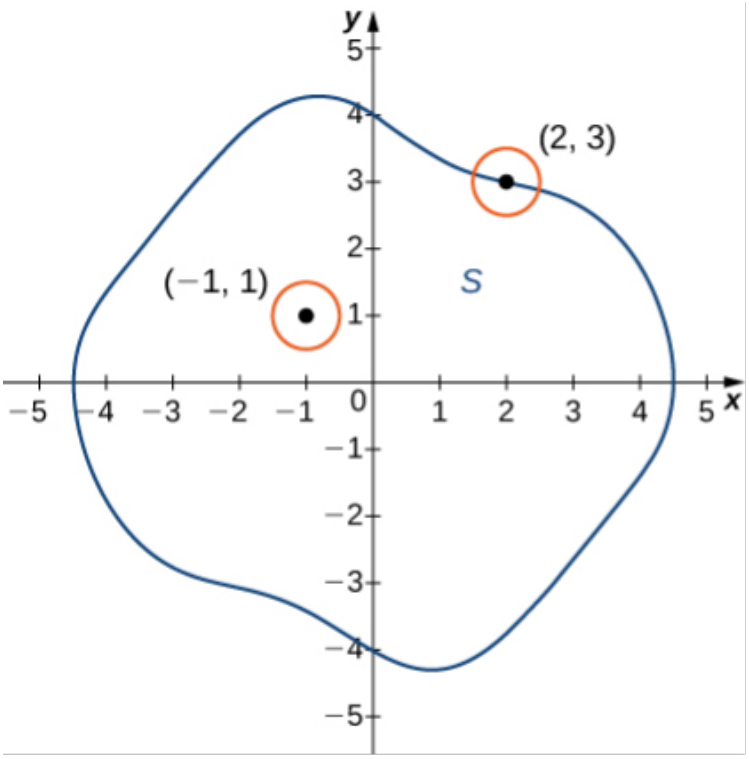


Figure. A planar set with one interior point and one boundary point marked.

Remark

Note that a boundary point of a set S *need not be contained in the set* S. For example, the δ-disk defined in this section does not contain its circular boundary. A set that contains its boundary is called *closed*.

Definition: Closed sets

A set $S \subset \mathbb{R}^n$ is called closed if it contains its boundary. A set $S \subset \mathbb{R}^n$ is bounded if it is contained in some ball centered at the origin, or equivalently if all points of S lie within some fixed distance of the origin.

Derivative reminders

The following table collects the general rules, trigonometric derivatives, and exponential and logarithmic derivatives. Here c is a constant, n is a real exponent, and f and g are differentiable functions. The formulas apply where the displayed expressions are defined and differentiable. Trigonometric arguments are measured in radians.

- **Expression:** General formulas
- **Expression:** c
  Derivative: 0
- **Expression:** x^n
  Derivative: nx^{n-1}
- **Expression:** $cf(x)$
  Derivative: $cf'(x)$
- **Expression:** $f(x) + g(x)$
  Derivative: $f'(x) + g'(x)$
- **Expression:** $f(x) - g(x)$
  Derivative: $f'(x) - g'(x)$

- **Expression:** $f(x)g(x)$
  Derivative: $f'(x)g(x) + f(x)g'(x)$
- **Expression:** $\dfrac{f(x)}{g(x)}$
  Derivative: $\dfrac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$
- **Expression:** $f(g(x))$
  Derivative: $f'(g(x))g'(x)$
- **Expression:** Trigonometric functions
- **Expression:** $\sin x$
  Derivative: $\cos x$
- **Expression:** $\cos x$
  Derivative: $-\sin x$
- **Expression:** $\tan x$
  Derivative: $\sec^2 x$
- **Expression:** $\cot x$
  Derivative: $-\csc^2 x$
- **Expression:** Exponential and logarithmic functions
- **Expression:** e^x
  Derivative: e^x
- **Expression:** b^x
  Derivative: $b^x \ln b$
- **Expression:** $\ln x$
  Derivative: $\dfrac{1}{x}$
- **Expression:** $\log_b x$
  Derivative: $\dfrac{1}{x \ln b}$

In the quotient rule, $g(x) \neq 0$. For the exponential and logarithmic formulas, $b > 0$ and $b \neq 1$. The formulas for $\ln x$ and $\log_b x$ require $x > 0$.